Strong disorder for 2D directed polymers and stochastic heat flow

Francesco Caravenna
University of Milano-Bicocca

Localization, Delocalization, and Diffusion in Disordered Systems
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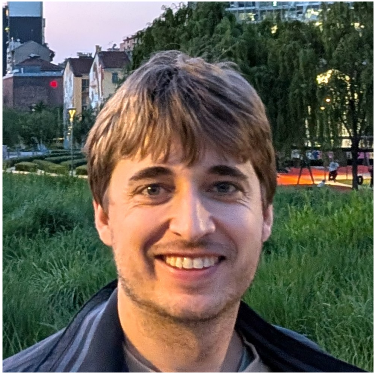
Collaborators

Quentin Berger
(Sorbonne Paris Nord)



Nicola Turchi
(Milano-Bicocca)

Collaborators

Rongfeng Sun (NUS)

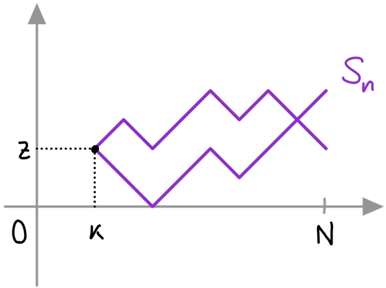
and

Nikos Zygouras (Warwick)

Outline

Directed polymer in random environment

- $S = (S_n)_{n\geq 0}$ simple random walk on $\mathbb{Z}^d$



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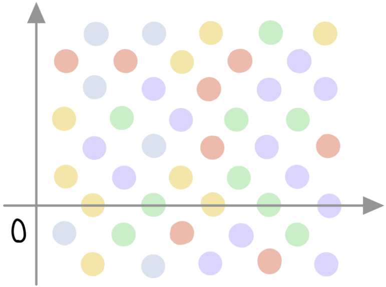

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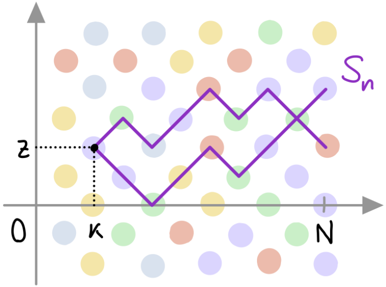

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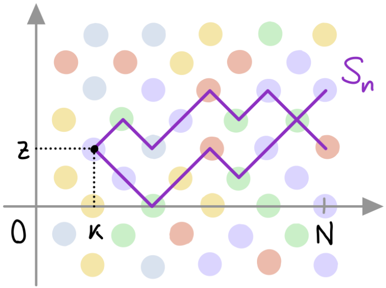

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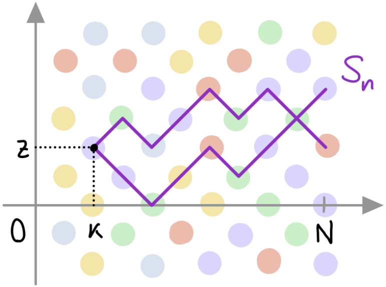

Partition Functions $(k \in \mathbb{N},\ z \in \mathbb{Z}^d)$

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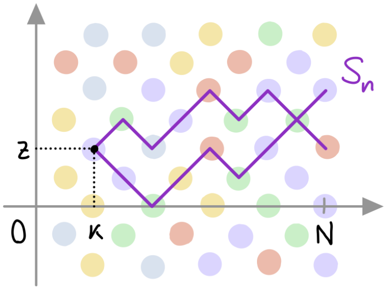

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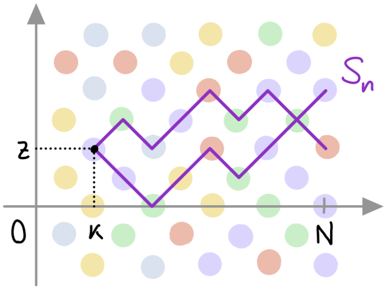

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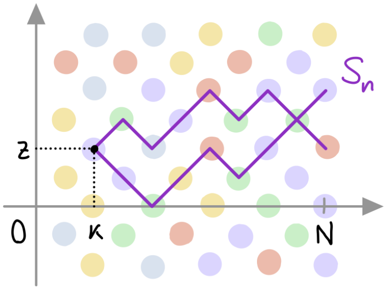

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Partition functions and SHE

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$$\lim_{N\to\infty} u_N(t,x)\,\mathrm{d}x = \mathscr{Z}_t^{\vartheta}(\mathrm{d}x) \qquad \text{renomalizing } \beta = \beta_N^{\text{crit}} \to 0$$

Deterministic Heat Equation (by Q. Berger)

Stochastic Heat Equation: subcritical (by Q. Berger)

Stochastic Heat Equation: critical (by Q. Berger)

Stochastic Heat Equation: supercritical (by Q. Berger)

From directed polymer to SHF

Averaged partition functions

$$Z_{N,\beta}(f) := \sum_{z\in\mathbb{Z}^2} f(z)\, Z_{N,\beta}(k=0, z)$$

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Averaged partition functions on balls

$$Z_{N,\beta}(f) := \sum_{z\in\mathbb{Z}^2} f(z)\, Z_{N,\beta}(k=0,z) \qquad\qquad f(\cdot) = B_R(\cdot) = \frac{1}{\pi R^2}\mathbb{1}_{|\cdot|\leq R}$$

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Convergence to Stochastic Heat Flow

$$Z_{N,\beta_N^{\mathrm{crit}}}(B_{\sqrt{N}}) \xrightarrow[N\to\infty]{d} \mathscr{Z}_t^{\vartheta}(B_1)$$

Outline

Beyond the critical regime

Critical regime $\beta = \beta_N^{\text{crit}}$:

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One-to-one correspondence (given N)

$$\beta \leftrightarrow \vartheta$$

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Supercritical directed polymer

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Main Theorem [Berger C. Turchi 25+]

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LB: 2^{nd} moment method

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Tail bounds and free energy

Right tail bounds

for $\varepsilon \in (0,1)$

$$\mathbb{P}\Big(Z_{N,\beta}\big(B_{\sqrt{N}}\big) \geq \varepsilon\Big)$$

Tail bounds and free energy

Right tail bounds for $\varepsilon \in (0,1)$

$$c_\varepsilon \exp(-C\, \mathrm{e}^{\vartheta}) \;\leq\; \mathbb{P}\Big(Z_{N,\beta}(B_{\sqrt{N}}) \geq \varepsilon\Big) \;\leq\; C_\varepsilon \exp(-c\, \mathrm{e}^{\vartheta})$$

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Free energy (point-to-plane)

$$\mathrm{F}(\beta) \;:=\; \lim_{N\to\infty} \frac{1}{N} \log Z_{N,\beta}(0)$$

Tail bounds and free energy

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Free energy (point-to-plane) [Berger Lacoin 17]

$$\mathrm{F}(\beta) := \lim_{N\to\infty} \frac{1}{N} \log Z_{N,\beta}(0) = \exp\Big(-\frac{\pi + o(1)}{\beta^2}\Big) \qquad \text{as } \beta \downarrow 0$$

Tail bounds and free energy

Right tail bounds

for $\varepsilon \in (0,1)$

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Back to discretized SHE

Solution of discretized SHE

$$u_{N,\beta}(t,x) = Z_{N,\beta}\left(N(1-t), \sqrt{N}x\right)$$

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Local extinction for β “$\gg$” β_N^{crit} $\qquad$ (e.g. for fixed $\beta > 0$)

Superdiffusivity of the supercritical SHE

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Outline

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Analogous results for the SHF $\mathscr{Z}_t^{\vartheta}$

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Quantitative bounds

Theorem [Berger C. Turchi 25+]

$$c \exp(-C\, t\, \mathrm{e}^{\vartheta}) \;\leq\; \mathbb{E}\big[\mathscr{Z}_t^{\vartheta}(B_{\sqrt{t}}) \wedge 1\big] \;\leq\; C \exp(-c\, t\, \mathrm{e}^{\vartheta}) \qquad \forall t, \vartheta$$

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LB agrees with log-log fluctuations of $\mathscr{Z}_t^{\vartheta}(B_r)$ as $r \downarrow 0$ [Gu Tsai 26+]

Outline

Proof of the UB: coarse-graining

$$\mathbb{E}\Big[Z_{N,\beta}\big(B_{e^{\hat{c}\, e^{\vartheta}}\sqrt{N}}\big) \wedge 1\Big] \;\le\; C\,\exp(-c\,e^{\vartheta})$$

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Key bound

$$\mathbb{E}\Big[Z_{N,\beta}\big(B_{\sqrt{N}}\big)\wedge 1\Big] \;\le\; \frac{C}{\vartheta} \qquad \text{for fixed } \vartheta \in \mathbb{R}$$

Proof of the UB: change of measure

Change of scale: $\mathbb{E}\Big[Z_{N,\beta}\big(B_{\sqrt{N}}\big) \wedge 1\Big] \le \frac{C}{\varepsilon}\, \mathbb{E}\Big[Z_{N,\beta}\big(B_{\sqrt{\varepsilon N}}\big) \wedge 1\Big] \qquad \forall \varepsilon \in (0,1)$

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Optimal choice $A = \{Z > 1\}$ (but we don't know Z)

Proof of the UB: choice of a proxy

Take $A = \{X > \frac{1}{2}\tilde{\mathbb{E}}[X]\}$ for some r.v. X with $\mathbb{E}[X] = 0$ $\qquad (X \approx Z - 1)$

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We take X as a manageable proxy of Z restricting its chaos expansion to

$$I = \{(n_1, x_1), \dots, (n_k, x_k)\} \qquad \text{with} \quad \begin{cases} \mathrm{width}(I) = n_k - n_1 \leq \varepsilon N, \\ \end{cases}$$

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By Markov and Chebychev $\qquad \mathbb{P}(A) \le 4\,\dfrac{\mathbb{V}\mathrm{ar}[X]}{\tilde{\mathbb{E}}[X]^2} \qquad \tilde{\mathbb{P}}(A^c) \le 4\,\dfrac{\tilde{\mathbb{V}}\mathrm{ar}[X]}{\tilde{\mathbb{E}}[X]^2}$

We take X as a manageable proxy of Z restricting its chaos expansion to

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We finally estimate $\mathbb{V}\mathrm{ar}[X]$, $\tilde{\mathbb{E}}[X]$ (2^{nd} moment) and $\tilde{\mathbb{V}}\mathrm{ar}[X]$ (3^{rd} moment)

Outline

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large time and/or strong disorder

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Robust proof based on coarse-graining + change of scale + change of measure

Thanks

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- Conditional GMC on path space $C([0,\infty), \mathscr{M}(\mathbb{R}^2))$ [Clark–Tsai 25+]